The H-join operation on signed graphs

Callum Huntington

University of Naples Federico II, Italy

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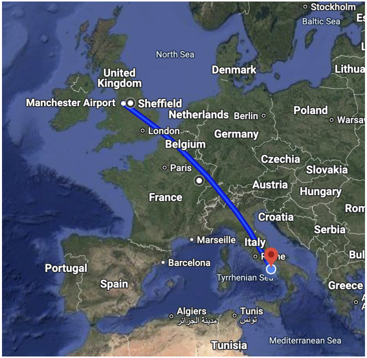
I am Callum Huntington. Originally from Sheffield in England, where I completed my BSc and MSc degrees, I moved to Italy in 2023 to begin my PhD at University of Naples Federico II, studying spectral graph theory with Professor Francesco Belardo.



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Some basics on spectral graph theory

Terminology and notation

In graph theory, a graph is typically represented as $G = (V, E)$, where V is the set of vertices (sometimes called nodes or points) and E is the set of edges (sometimes called arcs or links or lines).

Two vertices $x, y \in V$ are considered adjacent if they are connected by an edge $xy \in E$. An edge is said to be incident with (or to) the vertices it connects.

In undirected graphs, an edge xy is the same as yx, the order of the vertices in the edge does not matter. In directed graphs however, the order is important.

In a simple graph, there is at most one edge between any two vertices, and no loops (edges connecting a vertex to itself).

Signed graphs

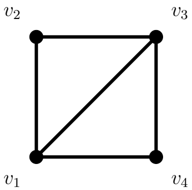
A signed graph $\dot{G}$ is an ordered pair $(G, \sigma) = G_\sigma$ where $G = (V, E)$ is an ordinary unsigned graph (the underlying graph) and $\sigma : E \longrightarrow \{+1, -1\}$ is the signature map on the edges of G. In general the underlying graph G has no restrictions, however we shall be considering finite simple (no loops or multiple edges) graphs.

Most of the notation used for simple graphs is also valid for signed graphs, regardless of the signature. For example:

- order (number of vertices) and size (number of edges),
- degree (total number of incident edges at each vertex),
- diameter (length of the longest shortest path between any two vertices).

However there are definitions which do depend on the signature. In signed graphs we have positive and negative walks, net-degrees, and the important property of being balanced.

Adjacency matrix



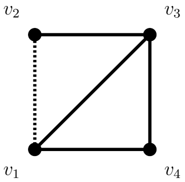
$$A_G = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

An adjacency matrix is a square matrix used to represent a finite graph. The elements of the matrix indicate whether pairs of vertices are adjacent or not in the graph.

We define the adjacency matrix A_G of a graph G with n vertices as the $n \times n$ matrix

$$(a_{ij}) = \begin{cases} 1, & ij \in E, \\ 0, & ij \notin E. \end{cases}$$

Adjacency matrix of a signed graph



$$A_{\dot{G}} = \begin{pmatrix} 0 & -1 & 1 & 1 \\ -1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

The adjacency matrix of a signed graph is very much like that of a standard graph, except now we have a -1 in each position corresponding to a negative edge. In the diagram here and throughout, a dashed line represents a negative edge.

Thus we define the adjacency matrix $A_{\dot{G}}$ of a signed graph $\dot{G}$ with n vertices as the $n \times n$ matrix

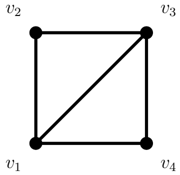
$$(a_{ij}) = \begin{cases} \sigma(ij), & ij \in E, \\ 0, & ij \notin E. \end{cases}$$

Laplacian matrix

For a vertex $v \in V$, define the set of its neighbours as

$$N(v) := \{u \in V : uv \in E\}.$$

The degree of v is $d_v = |N(v)|$, the number of neighbours v has and the degree matrix D_G of G is the $n \times n$ diagonal matrix with entries $d_1, d_2, \ldots, d_n$.



$$L_G = \begin{pmatrix} 3 & -1 & -1 & -1 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ -1 & 0 & -1 & 2 \end{pmatrix}$$

Then, the Laplacian matrix L_G of G is the matrix $L_G = D_G - A_G$.

Spectral graph theory

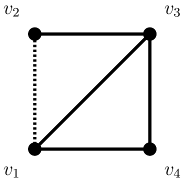
Spectral graph theory makes use of linear algebra to find correlations between combinatorial graph invariants and spectral invariants coming from related graph matrices.

When X is a given graph matrix associated with G, the X-spectrum of G consists of the eigenvalues of this matrix X.

It is important to note that the spectrum of a graph matrix does not depend on the labelling of the vertices. Two graphs which are the same up to labelling are called isomorphic, and their matrices can be obtained from one another by multiplication of permutation matrices. On the other hand, the eigenvectors will be different.

In the examples we have been using, we have the adjacency spectrum of G as $\{2.561, 0, -1, -1.561\}$ and the adjacency spectrum of $\dot{G}$ as $\{2, 1, -1, -2\}$.

Spectral properties



$$A_{\dot{G}} = \begin{pmatrix} 0 & -1 & 1 & 1 \\ -1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

Some properties of simple graphs follow through to the theory of signed graphs but not all.

For example, a simple graph has symmetric spectrum with respect to the origin (that is, λ is an eigenvalue if and only if $-\lambda$ is an eigenvalue, with the same multiplicity) if and only if the graph is bipartite. However, as we have already seen, there are nonbipartite signed graphs which do have symmetric spectrum.

Here the signed graph is clearly nonbipartite - we have cycles of odd length - yet the adjacency spectrum is $\{2, 1, -1, -2\}$.

Structural properties

From the A-eigenvalues of a signed graph we can deduce:

- the number of vertices and edges,
- the difference between the number of positive and negative triangles (that is, $\frac{1}{6}\sum \lambda_i^3$),
- the difference between the number of positive and negative closed walks of length p (that is, $\sum \lambda_i^p$).

From the L-eigenvalues of a signed graph we can deduce:

- the number of vertices and edges,
- the number of balanced components,
- the quantity $\sum d_i^2$,
- the quantity $\sum d_i^3 + 6(t^- - t^+)$, where t^- and t^+ are the number of negative and positive triangles, respectively.

Some more graph matrices

The net-degree $d_i^{\pm} = d_{\dot{G}}^{\pm}(v_i)$ of a vertex v_i in a signed graph $\dot{G}$ is defined to be $d_{\dot{G}}^{\pm}(v_i) = d_{\dot{G}}^{+}(v_i) - d_{\dot{G}}^{-}(v_i)$.

We define the net-degree matrix $D_{\dot{G}}^{\pm}$ in the natural way, as the $n \times n$ diagonal matrix with diagonal entries $d_1^{\pm}, d_2^{\pm}, \ldots, d_n^{\pm}$.

Hence the net-Laplacian matrix of a signed graph $\dot{G} = (G, \sigma)$, denoted by $L_{\dot{G}}^{\pm}$, is defined by $L_{\dot{G}}^{\pm} = D_{\dot{G}}^{\pm} - A_{\dot{G}}$.

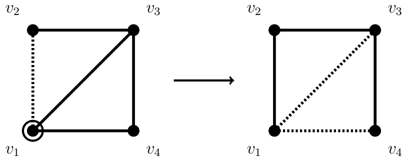
If we define $D_G^{-1/2}$ to be the matrix $\mathrm{diag}(1/\sqrt{d_1}, \ldots, 1/\sqrt{d_n})$ then the normalised adjacency and normalised Laplacian matrices are $\mathcal{A}_G = D_G^{-1/2} A_G D_G^{-1/2}$ and $\mathcal{L}_G = D_G^{-1/2} L_G D_G^{-1/2}$, respectively.

The eigenvalues of $\mathcal{A}$ lie in the interval $[-1, 1]$ and the eigenvalues of $\mathcal{L}$ lie in the interval $[0, 2]$.

Switching

A key aspect in signed graph theory is the idea of switching. This means reversing the signs of all edges between a vertex subset and its complement.

Shown below is a switching on v_1. Note that switching on multiple vertices is the same as switching on each vertex one step at a time.

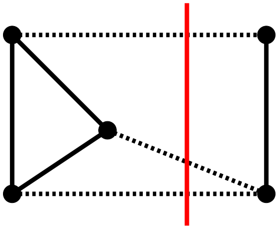


Two signed graphs $\dot{G}$ and $\dot{G}'$ on the same vertex set are switching equivalent if and only if their adjacency matrices satisfy $A_{\dot{G}} = SA_{\dot{G}'}S^{-1}$ for some diagonal matrix S which has entries of either $+1$ or -1.

Since there is no change made to the spectrum of a matrix after it has been conjugated, any two switching equivalent signed graphs have exactly the same characteristic polynomials and eigenvalues.

Balance

A signed graph is said to be balanced if the product of edge signs around every cycle is positive. It is well known that a signed graph is balanced if and only if it is switching equivalent to the all-positive signed graph.



The first characterisation of balance came from Harary in 1953. He said that a signed graph is balanced if and only if its vertex set can be divided into two subsets (either of which may be empty), so that each edge between the subsets is negative, and each edge within a subset is positive.

Balance from the spectrum

Balance in a signed graph can be recognised from the adjacency matrix and its spectrum. Stanić proved that if the largest A-eigenvalue of $\dot{G}$ is equal to the largest A-eigenvalue of the underlying graph G, then the graph is balanced.

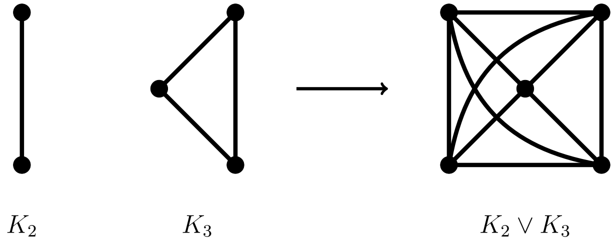
As mentioned before, we can also tell how many balanced components there are in a signed graph by looking at its Laplacian spectrum. For $\dot{G}$, the multiplicity of 0 appearing as an L-eigenvalue is equal to the number of balanced components in $\dot{G}$.

The H-join

Standard join

Usually, considering two graphs G_1 and G_2, the join $G_1 \vee G_2$ is taken to be the union of the existing vertices and edges along with new edges connecting each vertex of G_1 to every vertex of G_2.

Here we have K_2, K_3, and the join $K_2 \vee K_3$.



Symbolically it is the graph G such that $V_G = V_{G_1} \cup V_{G_2}$ and $E_G = E_{G_1} \cup E_{G_2} \cup \{uv : u \in V_{G_1}, v \in V_{G_2}\}$.

H-join

A generalisation of this operation works in the following way.

- Consider a graph H of order k and a family of graphs $\mathcal{F} = \{G_1, \ldots, G_k\}$.
- Replace each vertex v_i of H with the graph G_i.
- Connect each vertex of G_i to every vertex of G_j when v_i and v_j are adjacent in H.

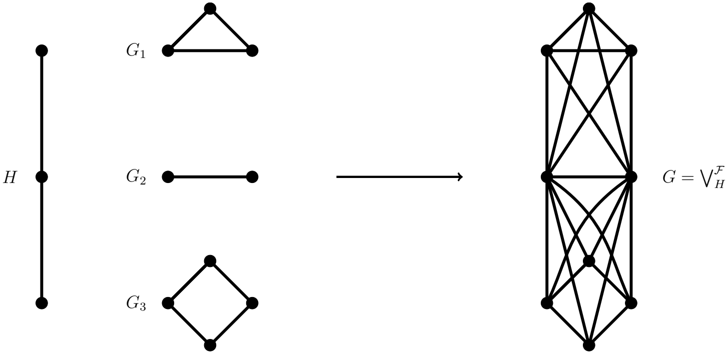
Hence, when $H = K_2$, the H-join corresponds precisely to the usual join graph operation.

Symbolically it is the graph $G = \bigvee_H^{\mathcal{F}}$ such that $V_G = \bigcup_{i=1}^k V_{G_i}$ and

$$E_G = \left(\bigcup_{i=1}^{k} E_{G_i} \right) \cup \left(\bigcup_{ij \in E_H} \{uv : u \in V_{G_i}, v \in V_{G_j}\} \right).$$

H-join

Here we have the H-join of $\mathcal{F} = \{K_3, K_2, C_4\}$ with $H = P_3$.



H-join history with simple graphs

1974: The first iteration of the operation we are studying was introduced as the generalised composition of graphs by Schwenk.[1]

2011: Cardoso et al. found the adjacency spectra of the H-join of a family of regular graphs with H being restricted to a path graph.[2]

2013: Cardoso et al. reintroduced the procedure and gave it the name of H-join. They also derived the spectra of H-joins of a family of regular graphs but this time with no restriction on H.[3]

2013: A variation known as the H-generalised join was presented by Cardoso et al. Here the adjacencies between component graphs are restricted according to vertex subsets.[4]

H-join history with simple graphs

2014: Both the signless Laplacian ($Q_G = D_G + A_G$) and the normalised Laplacian ($\mathcal{L}_G = D_G^{-1/2} L_G D_G^{-1/2}$) spectra of H-joins were determined by Wu et al.[5]

2022: Cardoso et al. were able to characterise the universal adjacency ($U_G = \alpha A_G + \beta I + \gamma J + \delta D_G$, with $\alpha, \beta, \gamma, \delta \in \mathbb{R}$, and $\alpha \neq 0$) spectrum of a graph obtained by the H-join operation, now to a family of arbitrary graphs.[6]

2024: More recently, Arunkumar and Ganeshbabu proposed a further generalisation of this operation, the H_m-join, this time constraining adjacencies according to indexing maps on the vertices of the component graphs.[7]

H-join history with signed graphs

2019: Brunetti et al. computed both the adjacency and Laplacian spectra of signed graphs obtained from the lexicographic product operation, something similar to the H-join but more limited.[8]

2021: For the $\dot{H}$-join proper on signed graphs, the operation was introduced and the adjacency and Laplacian spectra characterised by Zhang et al.[9]

2023: Li et al. produced signed graphs by performing the $\dot{H}$- and $\dot{H}$-generalised join operations and were able to determine the characteristic polynomials and spectra of many related matrices.[10]

2025: I have worked on the signed $\dot{H}_m$-join and the spectra of many of its matrices, this paper has been submitted and is under review.[11]

H-join adjacency matrix

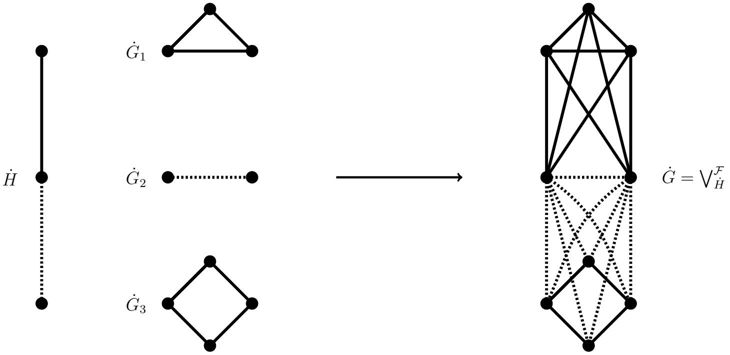
The adjacency matrix of the H-join is essentially obtained by replacing each entry of A_H with a block matrix. Along the diagonal we put the adjacency matrices of the G_i, where there is a 1 we substitute in an all-one matrix of the relevant size, and where there is a 0 we substitute in an all-zero matrix.

Let G be the H-join of $\mathcal{F} = \{K_3, K_2, C_4\}$ with $H = P_3$ from before. Then we have

$$A_H = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad A_G = \left(\begin{array}{ccc|cc|cccc} 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ \hline 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ \hline 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 \end{array}\right).$$

$\dot{H}$-join of signed graphs

The signed variant of this operation works in a similar way, only now we have to make the joins with respect the signature of the edges in $\dot{H}$. In the adjacency matrix too, we may now have blocks of -1 as well as 1 and 0.



$\dot{H}$-join balancedness

Zhang et al. show the following:

Let $\dot{H}$ be a signed graph of order k with no isolated vertex and let $\mathcal{F} = \{\dot{G}_1, \ldots, \dot{G}_k\}$ be a family of graphs such that $\bigcup_{i=1}^{k} E_{\dot{G}_i} \neq \emptyset$. If $\dot{G} = \vee_{\dot{H}}^{\mathcal{F}}$, then $\dot{G}$ is balanced if and only if $\dot{H}$ is balanced and each $\dot{G}_i$ is all-positive.

If $\dot{H} = (H, \sigma_H)$ is balanced, it is switching equivalent to the all-positive signed graph $(H, +)$. From here it is fairly easy to see that $\dot{G}$ is switching equivalent to the all-positive graph - we will look at this again later on - and so it is balanced.

On the other hand, if there is a negative edge in any $\dot{G}_i$, making any connection to a vertex in $\dot{G}_j$ $(i \neq j)$ will create an unbalanced triangle. Hence $\dot{G}$ is unbalanced.

The H-generalised and H_m-joins

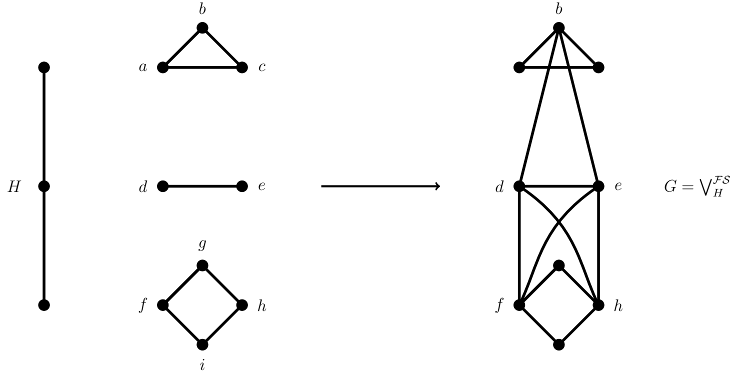
H-generalised join

The H-generalised join was introduced in 2013 by Cardoso et al. The main idea is similar, but this time rather than making joins between all the vertices of the component graphs, only a select subset of their vertices are used.

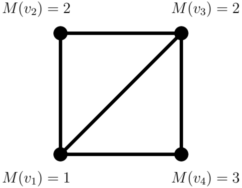
- Consider a graph H of order k and a family of graphs $\mathcal{F} = \{G_1, \ldots, G_k\}$.
- Consider a family of vertex subsets $\mathcal{S} = \{S_1, \ldots, S_k\}$, such that $S_i \subseteq V_{G_i}$.
- We substitute in each graph G_i for each vertex v_i of H and then make the joins between the vertices in the S_is where the edges exist in H.

H-generalised join

Here we have the H-generalised join of $\mathcal{F} = \{K_3, K_2, C_4\}$ with $H = P_3$ and $\mathcal{S} = \{\{b\}, \{d, e\}, \{f, h\}\}$.



Indexing maps



$$E = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

A further generalisation is known as the H_m-join, and in this situation the operation is constrained by indexing maps.

For the vertices of each G_i we have a map M_i, such that $M_i : V_{G_i} \to \{1, \ldots, m\}$. The m in the H_m-join refers to this largest value that a vertex can be sent to by the indexing maps.

The corresponding indexing matrices E_i are defined so that $(E)_{st} = 1$ if $M(v_s) = t$ and $(E)_{st} = 0$ otherwise. Hopefully this is a little easier to understand by looking at the diagram above.

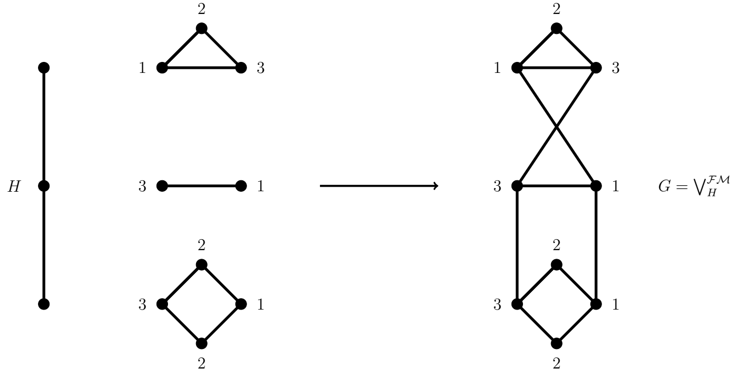
H_m-join

The H_m-join was introduced in 2024 by Arunkumar and Ganeshbabu. The idea is again similar, but this time rather than making joins between all the vertices within given subsets, each vertex has an index value, and then those with equal values that are in adjacent graphs are joined.

- Consider a graph H of order k and a family of graphs $\mathcal{F} = \{G_1, \ldots, G_k\}$.
- Consider a family of index maps $\mathcal{M} = \{M_1, \ldots, M_k\}$, such that $M_i : V_{G_i} \to \{1, \ldots, m\}$.
- We substitute in each graph G_i for each vertex v_i of H and then make the joins between the vertices in the G_is where the edges exist in H and the vertices have equal index values.

H_m-join

Here we have the H_m-join of $\mathcal{F} = \{K_3, K_2, C_4\}$ where $H = P_3$ with respect to the indexing maps M_1, M_2, M_3 whose values are labelled on the vertices.



$H \to H$-generalised $\to H_m$

We can see that the H-join operation is the H-generalised operation where $S_i = V_{G_i}$ in all cases, and is the H_m-join operation where $m = 1$ and $M : V_{G_i} \to \{1\}$.

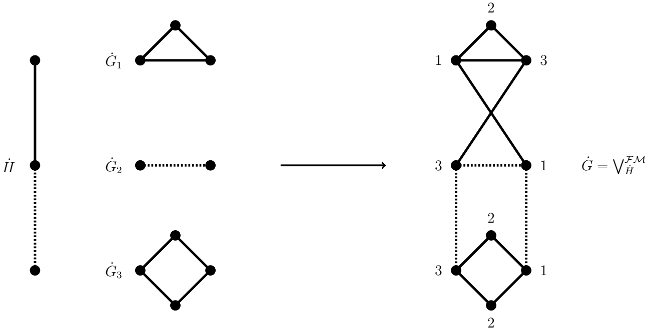
Consider an H-generalised join with H as a graph of order k, and let $m = k + 1$. For each $i \in \{1, \ldots, k\}$ we can define $M_i(v) = 1$ if $v \in S_i$ and $M_i(v) = i + 1$ otherwise. Then we can see that the H-generalised operation is an H_m-join under these conditions.

Various graph families can be realised as H_m-joins. For instance generalised Petersen graphs, wheel graphs, lollipop graphs, tadpole graphs, Helm graphs, and so on.

$\dot{H}_m$-join

As before, the signed variant of this operation works in a similar way. We simply make the joins with respect the signature of the edges in H.

Here we have the $\dot{H}_m$-join of $\mathcal{F} = \{K_3, -K_2, C_4\}$ where $H = \dot{P}_3$ with respect to the indexing maps M_1, M_2, M_3 whose values are labelled on the vertices.



$\dot{H}_m$-join adjacency matrix

Let $\dot{G} = \bigvee_{\dot{H}}^{\mathcal{F}\mathcal{M}}$, let $A_{\dot{G}_i}$ be the adjacency matrix of the signed graph $\dot{G}_i$, and let

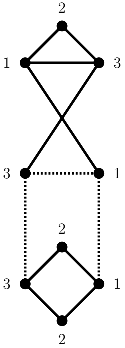
$$\omega_{i,j} = \begin{cases} \sigma_H(v_i v_j) & \text{if } v_i v_j \in E(H), \\ 0 & \text{otherwise.} \end{cases}$$

Let E_i be the $n_i \times m$ indexing matrix of $\dot{G}_i$ for the map M_i for all $1 \leq i \leq k$. Then, when suitably ordering the vertices, the adjacency matrix of $\dot{G}$ is

$$A_{\dot{G}} = \begin{pmatrix} A_{\dot{G}_1} & \omega_{1,2} E_1 E_2^T & \cdots & \omega_{1,k} E_1 E_k^T \\ \omega_{2,1} E_2 E_1^T & A_{\dot{G}_2} & \cdots & \omega_{2,k} E_2 E_k^T \\ \vdots & \vdots & \ddots & \vdots \\ \omega_{k,1} E_k E_1^T & \omega_{k,2} E_k E_2^T & \cdots & A_{\dot{G}_k} \end{pmatrix}.$$

$\dot{H}_m$-join adjacency matrix

Following our example, we have $\omega_{1,2} = 1$ and $\omega_{2,3} = -1$. We have the following three indexing matrices.



$$E_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$E_2 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

$$E_3 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

Thus the adjacency matrix is

$$A_{\dot{G}} = \left(\begin{array}{ccc|cc|cccc} 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 1 & 0 & -1 & -1 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 & 0 & 0 & 0 & -1 & 0 \\ \hline 0 & 0 & 0 & -1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \end{array}\right).$$

$\dot{H}_m$-join adjacency spectrum

Let $\dot{G} = \bigvee_{\dot{H}}^{\mathcal{F}\mathcal{M}}$. Let $\phi_i = \det(\lambda I_{n_i} - A_{\dot{G}_i})$ be the characteristic polynomial of the adjacency matrix $A_{\dot{G}_i}$ for all $1 \leq i \leq k$. Let $n = \sum_{i=1}^{k} n_i$ and let $\Gamma_i = E_i^T(\lambda I_{n_i} - A_{\dot{G}_i})^{-1} E_i$.

Then

$$\det(\lambda I_n - A_{\dot{G}}) = \left(\prod_{i=1}^{k} \phi_i \right) \det(\tilde{A}),$$

where

$$\tilde{A} = \begin{pmatrix} I_m & -\omega_{1,2}\Gamma_1 & \cdots & -\omega_{1,k}\Gamma_1 \\ -\omega_{2,1}\Gamma_2 & I_m & \cdots & -\omega_{2,k}\Gamma_2 \\ \vdots & \vdots & \ddots & \vdots \\ -\omega_{k,1}\Gamma_k & -\omega_{k,2}\Gamma_k & \cdots & I_m \end{pmatrix}.$$

$\dot{H}_m$-join adjacency spectrum proof

In proving this theorem we mainly have to take care with a lot of matrix manipulation. Ultimately not too difficult, but too long to go into the details here.

Proof. From Definition 3.2 we have that the adjacency matrix of $\dot{G}$ is

$$A_{\dot{G}} = \begin{pmatrix} A_{\dot{G}_1} & \omega_{1,2}E_1E_2^T & \cdots & \omega_{1,k}E_1E_k^T \\ \omega_{2,1}E_2E_1^T & A_{\dot{G}_2} & \cdots & \omega_{2,k}E_2E_k^T \\ \vdots & \vdots & \ddots & \vdots \\ \omega_{k,1}E_kE_1^T & \omega_{k,2}E_kE_2^T & \cdots & A_{\dot{G}_k} \end{pmatrix}.$$

We shall prove this claim by way of induction on k. For $k = 2$, we can make use of Schur's complement formula from Lemma 2.1 and doing so gives the adjacency characteristic polynomial

$$\begin{aligned} \phi_{A_{\dot{G}}}(\lambda) &= \det\begin{pmatrix} \lambda I_{n_1} - A_{\dot{G}_1} & -\omega_{1,2}E_1E_2^T \\ -\omega_{2,1}E_2E_1^T & \lambda I_{n_2} - A_{\dot{G}_2} \end{pmatrix} \\ &= \det\left(\lambda I_{n_2} - A_{\dot{G}_2}\right)\det\left((\lambda I_{n_1} - A_{\dot{G}_1}) - \omega_{1,2}E_1E_2^T(\lambda I_{n_2} - A_{\dot{G}_2})^{-1}\omega_{2,1}E_2E_1^T\right) \\ &= \phi_2 \cdot \det\left((\lambda I_{n_1} - A_{\dot{G}_1}) - \omega_{1,2}\omega_{2,1}E_1\Gamma_2E_1^T\right). \end{aligned}$$

Applying Lemma 2.2(b) with $M = \lambda I_{n_1} - A_{\dot{G}_1}$, $X = E_1\Gamma_2$, and $Y = E_1$, we then have

$$\begin{aligned} \phi_{A_{\dot{G}}}(\lambda) &= \phi_2 \cdot \det\left(\lambda I_{n_1} - A_{\dot{G}_1}\right)\det\left(I_m - \omega_{1,2}\omega_{2,1}E_1^T(\lambda I_{n_1} - A_{\dot{G}_1})^{-1}E_1\Gamma_2\right) \\ &= \phi_1 \cdot \phi_2 \cdot \det(I_m - \omega_{1,2}\omega_{2,1}\Gamma_1\Gamma_2). \end{aligned}$$

In reversing Schur's complement formula we can see that this in fact gives

$$\det\left(\lambda I_n - A_{\dot{G}}\right) = \phi_1 \cdot \phi_2 \cdot \det\begin{pmatrix} I_m & -\omega_{1,2}\Gamma_1 \\ -\omega_{2,1}\Gamma_2 & I_m \end{pmatrix} = \left(\prod_{i=1}^{2}\phi_i\right)\det\left(\tilde{A}\right).$$

Hence the claim is true for $k = 2$.

Now for $k \geq 3$. By Schur's complement formula we obtain

$$\begin{aligned} \det\left(\lambda I_n - A_{\dot{G}}\right) &= \det\begin{pmatrix} \lambda I_{n_1} - A_{\dot{G}_1} & -\omega_{1,2}E_1E_2^T & \cdots & -\omega_{1,k}E_1E_k^T \\ -\omega_{2,1}E_2E_1^T & \lambda I_{n_2} - A_{\dot{G}_2} & \cdots & -\omega_{2,k}E_2E_k^T \\ \vdots & \vdots & \ddots & \vdots \\ -\omega_{k,1}E_kE_1^T & -\omega_{k,2}E_kE_2^T & \cdots & \lambda I_{n_k} - A_{\dot{G}_k} \end{pmatrix} \\ &= \det\left(\lambda I_{n_k} - A_{\dot{G}_k}\right)\det(S) \end{aligned} \tag{1}$$

where

$$\begin{aligned} S = &\begin{pmatrix} \lambda I_{n_1} - A_{\dot{G}_1} & -\omega_{1,2}E_1E_2^T & \cdots & -\omega_{1,k-1}E_1E_{k-1}^T \\ -\omega_{2,1}E_2E_1^T & \lambda I_{n_2} - A_{\dot{G}_2} & \cdots & -\omega_{2,k-1}E_2E_{k-1}^T \\ \vdots & \vdots & \ddots & \vdots \\ -\omega_{k-1,1}E_{k-1}E_1^T & -\omega_{k-1,2}E_{k-1}E_2^T & \cdots & \lambda I_{n_{k-1}} - A_{\dot{G}_{k-1}} \end{pmatrix} \\ &- \begin{pmatrix} -\omega_{1,k}E_1E_k^T \\ -\omega_{2,k}E_2E_k^T \\ \vdots \\ -\omega_{k-1,k}E_{k-1}E_k^T \end{pmatrix}(\lambda I_{n_k} - A_{\dot{G}_k})^{-1}\begin{pmatrix} -\omega_{k,1}E_kE_1^T & -\omega_{k,2}E_kE_2^T & \cdots & -\omega_{k,k-1}E_kE_{k-1}^T \end{pmatrix}. \end{aligned}$$

From here we shall compute the determinant of S and show that we acquire the characteristic polynomial which we desire. To begin with, for S we have

$$\begin{aligned} &= \begin{pmatrix} \lambda I_{n_1} - A_{\dot{G}_1} & -\omega_{1,2}E_1E_2^T & \cdots & -\omega_{1,k-1}E_1E_{k-1}^T \\ -\omega_{2,1}E_2E_1^T & \lambda I_{n_2} - A_{\dot{G}_2} & \cdots & -\omega_{2,k-1}E_2E_{k-1}^T \\ \vdots & \vdots & \ddots & \vdots \\ -\omega_{k-1,1}E_{k-1}E_1^T & -\omega_{k-1,2}E_{k-1}E_2^T & \cdots & \lambda I_{n_{k-1}} - A_{\dot{G}_{k-1}} \end{pmatrix} \\ &\quad - \begin{pmatrix} \omega_{1,k}\omega_{k,1}E_1\Gamma_kE_1^T & \omega_{1,k}\omega_{k,2}E_1\Gamma_kE_2^T & \cdots & \omega_{1,k}\omega_{k,k-1}E_1\Gamma_kE_{k-1}^T \\ \omega_{2,k}\omega_{k,1}E_2\Gamma_kE_1^T & \omega_{2,k}\omega_{k,2}E_2\Gamma_kE_2^T & \cdots & \omega_{2,k}\omega_{k,k-1}E_2\Gamma_kE_{k-1}^T \\ \vdots & \vdots & \ddots & \vdots \\ \omega_{k-1,k}\omega_{k,1}E_{k-1}\Gamma_kE_1^T & \omega_{k-1,k}\omega_{k,2}E_{k-1}\Gamma_kE_2^T & \cdots & \omega_{k-1,k}\omega_{k,k-1}E_{k-1}\Gamma_kE_{k-1}^T \end{pmatrix} \\ &= \begin{pmatrix} \lambda I_{n_1} - A_{\dot{G}_1} & O & \cdots & O \\ O & \lambda I_{n_2} - A_{\dot{G}_2} & \cdots & O \\ \vdots & \vdots & \ddots & \vdots \\ O & O & \cdots & \lambda I_{n_{k-1}} - A_{\dot{G}_{k-1}} \end{pmatrix} \\ &\quad - \begin{pmatrix} \omega_{1,k}\omega_{k,1}E_1\Gamma_kE_1^T & E_1(\omega_{1,2} + \omega_{1,k}\omega_{k,2}\Gamma_k)E_2^T & \cdots & E_1(\omega_{1,k-1} + \omega_{1,k}\omega_{k,k-1}\Gamma_k)E_{k-1}^T \\ E_2(\omega_{2,1} + \omega_{2,k}\omega_{k,1}\Gamma_k)E_1^T & \omega_{2,k}\omega_{k,2}E_2\Gamma_kE_2^T & \cdots & E_2(\omega_{2,k-1} + \omega_{2,k}\omega_{k,k-1}\Gamma_k)E_{k-1}^T \\ \vdots & \vdots & \ddots & \vdots \\ E_{k-1}(\omega_{k-1,1} + \omega_{k-1,k}\omega_{k,1}\Gamma_k)E_1^T & E_{k-1}(\omega_{k-1,2} + \omega_{k-1,k}\omega_{k,2}\Gamma_k)E_2^T & \cdots & \omega_{k-1,k}\omega_{k,k-1}E_{k-1}\Gamma_kE_{k-1}^T \end{pmatrix} \\ &= \begin{pmatrix} \lambda I_{n_1} - A_{\dot{G}_1} & O & \cdots & O \\ O & \lambda I_{n_2} - A_{\dot{G}_2} & \cdots & O \\ \vdots & \vdots & \ddots & \vdots \\ O & O & \cdots & \lambda I_{n_{k-1}} - A_{\dot{G}_{k-1}} \end{pmatrix} \\ &\quad - \left(\begin{pmatrix} \omega_{1,k}\omega_{k,1}E_1\Gamma_k & E_1(\omega_{1,2} + \omega_{1,k}\omega_{k,2}\Gamma_k) & \cdots & E_1(\omega_{1,k-1} + \omega_{1,k}\omega_{k,k-1}\Gamma_k) \\ E_2(\omega_{2,1} + \omega_{2,k}\omega_{k,1}\Gamma_k) & \omega_{2,k}\omega_{k,2}E_2\Gamma_k & \cdots & E_2(\omega_{2,k-1} + \omega_{2,k}\omega_{k,k-1}\Gamma_k) \\ \vdots & \vdots & \ddots & \vdots \\ E_{k-1}(\omega_{k-1,1} + \omega_{k-1,k}\omega_{k,1}\Gamma_k) & E_{k-1}(\omega_{k-1,2} + \omega_{k-1,k}\omega_{k,2}\Gamma_k) & \cdots & \omega_{k-1,k}\omega_{k,k-1}E_{k-1}\Gamma_k \end{pmatrix}\right. \\ &\qquad \left. \cdot \begin{pmatrix} E_1^T & O & \cdots & O \\ O & E_2^T & \cdots & O \\ \vdots & \vdots & \ddots & \vdots \\ O & O & \cdots & E_{k-1}^T \end{pmatrix}\right). \end{aligned}$$

Now we can apply Lemma 2.2(b) to obtain the determinant of S as

$$\begin{aligned} &= \left(\prod_{i=1}^{k}\phi_i\right) \cdot \det\left(I_{k(m-1)} - \begin{pmatrix} E_1^T & O & \cdots & O \\ O & E_2^T & \cdots & O \\ \vdots & \vdots & \ddots & \vdots \\ O & O & \cdots & E_{k-1}^T \end{pmatrix} \cdot \begin{pmatrix} \lambda I_{n_1} - A_{\dot{G}_1} & O & \cdots & O \\ O & \lambda I_{n_2} - A_{\dot{G}_2} & \cdots & O \\ \vdots & \vdots & \ddots & \vdots \\ O & O & \cdots & \lambda I_{n_{k-1}} - A_{\dot{G}_{k-1}} \end{pmatrix}^{-1}\right. \\ &\qquad \left. \cdot \begin{pmatrix} \omega_{1,k}\omega_{k,1}E_1\Gamma_k & E_1(\omega_{1,2} + \omega_{1,k}\omega_{k,2}\Gamma_k) & \cdots & E_1(\omega_{1,k-1} + \omega_{1,k}\omega_{k,k-1}\Gamma_k) \\ E_2(\omega_{2,1} + \omega_{2,k}\omega_{k,1}\Gamma_k) & \omega_{2,k}\omega_{k,2}E_2\Gamma_k & \cdots & E_2(\omega_{2,k-1} + \omega_{2,k}\omega_{k,k-1}\Gamma_k) \\ \vdots & \vdots & \ddots & \vdots \\ E_{k-1}(\omega_{k-1,1} + \omega_{k-1,k}\omega_{k,1}\Gamma_k) & E_{k-1}(\omega_{k-1,2} + \omega_{k-1,k}\omega_{k,2}\Gamma_k) & \cdots & \omega_{k-1,k}\omega_{k,k-1}E_{k-1}\Gamma_k \end{pmatrix}\right) \\ &= \left(\prod_{i=1}^{k}\phi_i\right) \cdot \det\left(I_{k(m-1)}\right. \\ &\qquad \left. - \begin{pmatrix} \omega_{1,k}\omega_{k,1}E_1\Gamma_k & \Gamma_1(\omega_{1,2} + \omega_{1,k}\omega_{k,2}\Gamma_k) & \cdots & \Gamma_1(\omega_{1,k-1} + \omega_{1,k}\omega_{k,k-1}\Gamma_k) \\ \Gamma_2(\omega_{2,1} + \omega_{2,k}\omega_{k,1}\Gamma_k) & \omega_{2,k}\omega_{k,2}E_2\Gamma_k & \cdots & \Gamma_2(\omega_{2,k-1} + \omega_{2,k}\omega_{k,k-1}\Gamma_k) \\ \vdots & \vdots & \ddots & \vdots \\ \Gamma_{k-1}(\omega_{k-1,1} + \omega_{k-1,k}\omega_{k,1}\Gamma_k) & \Gamma_{k-1}(\omega_{k-1,2} + \omega_{k-1,k}\omega_{k,2}\Gamma_k) & \cdots & \omega_{k-1,k}\omega_{k,k-1}E_{k-1}\Gamma_k \end{pmatrix}\right) \\ &= \left(\prod_{i=1}^{k}\phi_i\right) \cdot \det\left(\begin{pmatrix} I_m & -\omega_{1,2}\Gamma_1 & \cdots & -\omega_{1,k-1}\Gamma_1 \\ -\omega_{2,1}\Gamma_2 & I_m & \cdots & -\omega_{2,k-1}\Gamma_2 \\ \vdots & \vdots & \ddots & \vdots \\ -\omega_{k-1,1}\Gamma_{k-1} & -\omega_{k-1,2}\Gamma_{k-1} & \cdots & I_m \end{pmatrix}\right. \\ &\qquad \left. - \begin{pmatrix} \omega_{1,k}\omega_{k,1}\Gamma_1\Gamma_k & \omega_{1,k}\omega_{k,2}\Gamma_1\Gamma_k & \cdots & \omega_{1,k}\omega_{k,k-1}\Gamma_1\Gamma_k \\ \omega_{2,k}\omega_{k,1}\Gamma_2\Gamma_k & \omega_{2,k}\omega_{k,2}\Gamma_2\Gamma_k & \cdots & \omega_{2,k}\omega_{k,k-1}\Gamma_2\Gamma_k \\ \vdots & \vdots & \ddots & \vdots \\ \omega_{k-1,k}\omega_{k,1}\Gamma_{k-1}\Gamma_k & \omega_{k-1,k}\omega_{k,2}\Gamma_{k-1}\Gamma_k & \cdots & \omega_{k-1,k}\omega_{k,k-1}\Gamma_{k-1}\Gamma_k \end{pmatrix}\right) \\ &= \left(\prod_{i=1}^{k}\phi_i\right) \cdot \det\left(\begin{pmatrix} I_m & -\omega_{1,2}\Gamma_1 & \cdots & -\omega_{1,k-1}\Gamma_1 \\ -\omega_{2,1}\Gamma_2 & I_m & \cdots & -\omega_{2,k-1}\Gamma_2 \\ \vdots & \vdots & \ddots & \vdots \\ -\omega_{k-1,1}\Gamma_{k-1} & -\omega_{k-1,2}\Gamma_{k-1} & \cdots & I_m \end{pmatrix}\right. \\ &\qquad \left. - \begin{pmatrix} \omega_{1,k}\Gamma_1 \\ \omega_{2,k}\Gamma_2 \\ \vdots \\ \omega_{k-1,k}\Gamma_{k-1} \end{pmatrix}(\Gamma_k)\begin{pmatrix} \omega_{k,1} & \omega_{k,2} & \cdots & \omega_{k,k-1} \end{pmatrix}\right). \end{aligned}$$

Using Schur's complement formula once again we obtain

$$\det(S) = \left(\prod_{i=1}^{k}\phi_i\right) \cdot \det\begin{pmatrix} I_m & -\omega_{1,2}\Gamma_1 & \cdots & -\omega_{1,k}\Gamma_1 \\ -\omega_{2,1}\Gamma_2 & I_m & \cdots & -\omega_{2,k}\Gamma_2 \\ \vdots & \vdots & \ddots & \vdots \\ -\omega_{k,1}\Gamma_k & -\omega_{k,2}\Gamma_k & \cdots & I_m \end{pmatrix}.$$

Finally, from (1) we have

$$\det\left(\lambda I_n - A_{\dot{G}_i}\right) = \left(\prod_{i=1}^{k}\phi_i\right) \cdot \det\begin{pmatrix} I_m & -\omega_{1,2}\Gamma_1 & \cdots & -\omega_{1,k}\Gamma_1 \\ -\omega_{2,1}\Gamma_2 & I_m & \cdots & -\omega_{2,k}\Gamma_2 \\ \vdots & \vdots & \ddots & \vdots \\ -\omega_{k,1}\Gamma_k & -\omega_{k,2}\Gamma_k & \cdots & I_m \end{pmatrix} = \left(\prod_{i=1}^{k}\phi_i\right)\det\left(\tilde{A}\right).$$

This proves the theorem. $\square$

$\dot{H}_m$-join adjacency eigenvalues

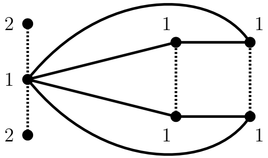
Let M be a square normal complex matrix of order n and let E be an $n \times m$ complex matrix. An eigenvalue λ of M is an E-main eigenvalue if its associated eigenspace $\xi_M(\lambda)$ is not orthogonal to the column space of E. Else, it is called an E-non-main eigenvalue.

Let λ be an eigenvalue of the signed graph $\dot{G}_i$ and let $\mu_i(\lambda)$ be its multiplicity.

- If λ is an E_i-non-main adjacency eigenvalue of a component graph $\dot{G}_i$ then λ is an adjacency eigenvalue of the join graph $\dot{G}$ with multiplicity no less than $\mu_i(\lambda)$.
- If λ is an E_i-main adjacency eigenvalue of a component graph $\dot{G}_i$ then λ is an adjacency eigenvalue of the join graph $\dot{G}$ with multiplicity no less than $\mu_i(\lambda) - m$.

A small example

Consider the $\dot{H}_m$-join presented below.



With a suitable ordering of the vertices we have the following:

$$E_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}, \quad \phi_1 = \lambda^3 - 2\lambda, \quad \Gamma_1 = E_1^T(\lambda I - A_{\dot{G}_1})^{-1}E_1 = \frac{1}{\lambda^2 - 2}\begin{pmatrix} \lambda & -2 \\ -2 & 2\lambda \end{pmatrix},$$

$$E_2 = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 0 \end{pmatrix}, \quad \phi_2 = \lambda^4 - 4\lambda^2, \quad \Gamma_2 = E_2^T(\lambda I - A_{\dot{G}_2})^{-1}E_2 = \frac{1}{\lambda}\begin{pmatrix} 4 & 0 \\ 0 & 0 \end{pmatrix}.$$

A small example

For $\dot{G}_1 = (P_3, -)$ the spectrum is $\{\sqrt{2}, 0, -\sqrt{2}\}$ and we have

$$\xi_{A_{\dot{G}_1}}(\sqrt{2}) = \mathrm{span}\left\{\begin{pmatrix}-\sqrt{2}\\1\\1\end{pmatrix}\right\},\quad \xi_{A_{\dot{G}_1}}(0) = \mathrm{span}\left\{\begin{pmatrix}0\\-1\\1\end{pmatrix}\right\},\quad \xi_{A_{\dot{G}_1}}(-\sqrt{2}) = \mathrm{span}\left\{\begin{pmatrix}\sqrt{2}\\1\\1\end{pmatrix}\right\}.$$

Hence 0 is an E_1-non-main eigenvalue and $\sqrt{2}$ and $-\sqrt{2}$ are E_1-main eigenvalues. This means that 0 will be an eigenvalue of $A_{\dot{G}}$ at least once.

For $\dot{G}_2 = (C_4, \sigma)$ the spectrum is $\{2, 0, 0, -2\}$ and we have

$$\xi_{A_{\dot{G}_2}}(2) = \mathrm{span}\left\{\begin{pmatrix}-1\\-1\\1\\1\end{pmatrix}\right\},\quad \xi_{A_{\dot{G}_2}}(0) = \mathrm{span}\left\{\begin{pmatrix}1\\0\\1\\0\end{pmatrix},\begin{pmatrix}0\\1\\0\\1\end{pmatrix}\right\},\quad \xi_{A_{\dot{G}_3}}(-2) = \mathrm{span}\left\{\begin{pmatrix}1\\-1\\-1\\1\end{pmatrix}\right\}.$$

Hence 2 and -2 are E_2-non-main eigenvalues and 0 is an E_2-main eigenvalue. This means that 2 and -2 will each be eigenvalues of $A_{\dot{G}}$ at least once.

A small example

By our theorem, we have

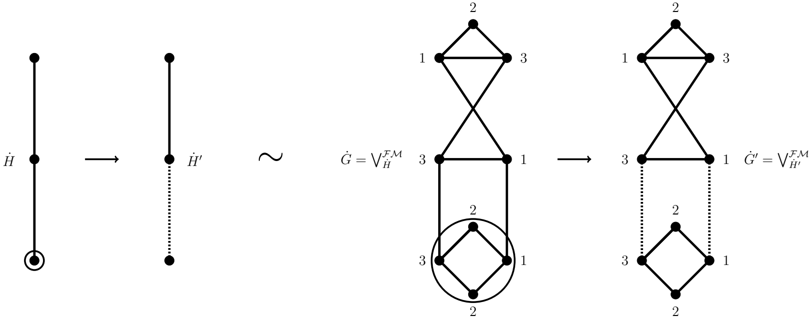
$$\begin{aligned}\det\left(\lambda I - A_{\dot{G}}\right) &= \phi_1 \cdot \phi_2 \cdot \det\begin{pmatrix} I_2 & -\Gamma_1 \\ -\Gamma_2 & I_2 \end{pmatrix} \\ &= (\lambda^3 - 2\lambda) \cdot (\lambda^4 - 4\lambda^2) \cdot \frac{\lambda^2 - 6}{\lambda^2 - 4} \\ &= \lambda^7 - 10\lambda^5 + 24\lambda^3.\end{aligned}$$

So the spectrum of $\dot{G}$ is $\{\sqrt{6}, 2, 0, 0, 0, -2, -\sqrt{6}\}$ and we can see that this satisfies the claims made about the E_i-non-main eigenvalues.

It can also be noted that this is another example of a signed graph which has a symmetric spectrum with respect to the origin but which is not bipartite.

Stability under switching of $\dot{H}$

Let $\dot{H} = (H, \sigma_H)$ and $\dot{H}' = (H, \sigma'_H)$ be switching equivalent. Then $\dot{G} = \bigvee_{\dot{H}}^{\mathcal{FM}}$ and $\dot{G}' = \bigvee_{\dot{H}'}^{\mathcal{FM}}$ are also switching equivalent.



Stability under switching of $\dot{H}$

Switching equivalence between $\dot{H}$ and $\dot{H}'$ implies that their adjacency matrices are switching similar. Hence for some switching matrix $S = (s_{ij})$ we have $SA_{\dot{H}}S^{-1} = A_{\dot{H}'}$.

Using this matrix to give the switching operation in terms of the signatures of $\dot{H}$ and $\dot{H}'$ we have $s_{ii}\sigma_H(v_iv_j)s_{jj} = \sigma'_H(v_iv_j)$. If $v_iv_j \notin E(H)$ then $\omega'_{i,j} = \omega_{i,j} = 0$ and if $v_iv_j \in E(H)$ we have

$$\omega'_{i,j} = \sigma'_H(v_iv_j) = s_{ii}\sigma_H(v_iv_j)s_{jj} = s_{ii}\omega_{i,j}s_{jj}.$$

Thus $\omega'_{i,j} = s_{ii}\omega_{i,j}s_{jj}$ for all $1 \le i, j \le k$.

Stability under switching of $\dot{H}$

Let $\dot{G} = \bigvee_{\dot{H}}^{\mathcal{FM}}$ and let $\dot{G}' = \bigvee_{\dot{H}'}^{\mathcal{FM}}$.

By considering the adjacency matrix of $\dot{G}'$ we can see that:

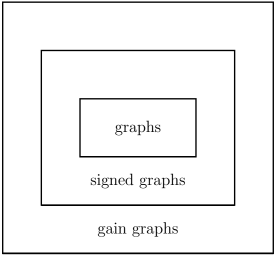
$$
\begin{aligned}
A_{\dot{G}'} &= \begin{pmatrix} A_{\dot{G}_1} & \omega'_{1,2}E_1E_2^T & \cdots & \omega'_{1,k}E_1E_k^T \\ \omega'_{2,1}E_2E_1^T & A_{\dot{G}_2} & \cdots & \omega'_{2,k}E_2E_k^T \\ \vdots & \vdots & \ddots & \vdots \\ \omega'_{k,1}E_kE_1^T & \omega'_{k,2}E_kE_2^T & \cdots & A_{\dot{G}_k} \end{pmatrix} \\
&= \begin{pmatrix} A_{\dot{G}_1} & s_{11}\omega_{1,2}s_{22}E_1E_2^T & \cdots & s_{11}\omega_{1,k}s_{kk}E_1E_k^T \\ s_{22}\omega_{2,1}s_{11}E_2E_1^T & A_{\dot{G}_2} & \cdots & s_{22}\omega_{2,k}s_{kk}E_2E_k^T \\ \vdots & \vdots & \ddots & \vdots \\ s_{kk}\omega_{k,1}s_{11}E_kE_1^T & s_{kk}\omega_{k,2}s_{22}E_kE_2^T & \cdots & A_{\dot{G}_k} \end{pmatrix} \\
&= \begin{pmatrix} s_{11}I_{n_1} & O & \cdots & O \\ O & s_{22}I_{n_2} & \cdots & O \\ \vdots & \vdots & \ddots & \vdots \\ O & O & \cdots & s_{kk}I_{n_k} \end{pmatrix} \begin{pmatrix} A_{\dot{G}_1} & \omega_{1,2}E_1E_2^T & \cdots & \omega_{1,k}E_1E_k^T \\ \omega_{2,1}E_2E_1^T & A_{\dot{G}_2} & \cdots & \omega_{2,k}E_2E_k^T \\ \vdots & \vdots & \ddots & \vdots \\ \omega_{k,1}E_kE_1^T & \omega_{k,2}E_kE_2^T & \cdots & A_{\dot{G}_k} \end{pmatrix} \begin{pmatrix} s_{11}I_{n_1} & O & \cdots & O \\ O & s_{22}I_{n_2} & \cdots & O \\ \vdots & \vdots & \ddots & \vdots \\ O & O & \cdots & s_{kk}I_{n_k} \end{pmatrix}^{-1}.
\end{aligned}
$$

Hence $A_{\dot{G}'} = SA_{\dot{G}}S^{-1}$ for some switching matrix S and so by definition $\dot{G} = \bigvee_{\dot{H}}^{\mathcal{FM}}$ is switching equivalent to $\dot{G}' = \bigvee_{\dot{H}'}^{\mathcal{FM}}$ and we are done.

Gain graphs, briefly

Gain graphs

A gain graph is a graph whose edges are labelled invertibly, or orientably, by elements of a group. This means that, if an edge e in one direction has label g (a group element), then in the other direction it has label g^{-1}.



It is clear to see that simple graphs are gain graphs with the trivial multiplicative group $\{1\}$ and that signed graphs are gain graphs with the multiplicative group $\{+1, -1\}$.

Gain balance and switching

Much like in the setting of signed graphs, notions of balance and switching are present in the theory of gain graphs.

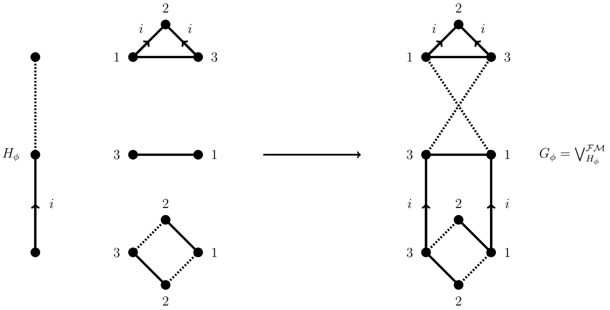
Two gain graphs are switching equivalent if and only if their adjacency matrices are conjugated by a diagonal matrix with diagonal entries in the group. This is analogous to the switching for signed graphs, where the entries were only $+1$ and -1.

A gain graph is said to be balanced if the product of edge signs around every cycle is equal to the identity element of the group. Again this resembles what we have for signed graphs, where the product of edge signs in cycles must be equal to $+1$ for balance.

Further results

It is expected (and hoped for!) that the research concerning the join operations discussed in this seminar can be extended to the field of gain graphs. Following similar steps, we just need to pay attention to the relevant gain functions ϕ_H and ϕ_{G_i}.

First considering the group of fourth roots of unity $\mathbb{T}_4 = \{\pm 1, \pm i\}$, we can have an example like so.



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Thank you!